Calorimetry lab

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use an energy balance to analyze mixing and heat transfer.

SUCCESS CRITERIA - I can...

1. Identify hot and cold samples and their masses.
2. Balance energy gained and lost in a simplified calculation.
3. Discuss heat exchange with the container and surroundings.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “calorimetry” and “isolated” to explain your answer. Sentence starters are optional.

calorimetry: using temperature changes to track energy transfers

isolated: exchanging no relevant quantity with surroundings in this model

heat: energy transferred because of a temperature difference

01 Warm-up: bring back an earlier idea

A **0.3 kg** sample absorbs **1500 J** while warming **5°C**. **Find specific heat** and **name one thing** that must be true for this calculation to work.

Word help: specific heat = energy per mass needed for a one-degree temperature change

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $Q_{hot} + Q_{cold} = \mathbf{0}$; $Q = mc(T_f - T_i)$.
Ideal isolated mixing, no phase change. Heat lost is negative; heat gained is positive. Real surroundings may exchange energy.

Worked reasoning: Two equal masses of the same liquid at **20°C** and **40°C** mix in an insulated container. The final **30°C** makes the hot sample lose the same energy the cool sample gains.

Your turn: Draw heat-transfer arrows and **show the two equal temperature** changes.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Equal masses of the same liquid at **20°C** and **60°C** are mixed in an ideal insulated container. **Predict the final temperature**. **Explain how you reached your** answer.

Word help: justify = **support your answer** with a reason or evidence

Sentence starter: I predict ... . I used ... to make this prediction because ... .

20°C

60°C

Each outline represents the labeled sample or container.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

You mix **0.10 kg** of water at **80°C** with **0.30 kg** of water at **20°C**. Ignore energy exchange with the cup and room. **Label both masses** and temperatures, predict which initial temperature the final value will be closer to, then **calculate equilibrium temperature**.

Sentence starter: I predict … . I used … to make this prediction because … .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

05 Explain your reasoning

Plan a table for masses, initial temperatures, final temperature and a heat-loss observation. Why should the final temperature be measured promptly after mixing?

Sentence starter: My design includes … . It does what is needed by … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A spring has k=**400 N/m** and stores **0.72 J**. **Find displacement magnitude** and force magnitude. **Mark equilibrium**, displacement and restoring-force direction.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.



Mark equilibrium and displacement x.

Marzano check - circle: 1 2 3

07 Exit check

A **0.20 kg** metal cools from **100** to **40°C** while **0.10 kg** water warms from **20** to **40°C**. Use water c=**4200**. Neglect cup and losses; **find metal c**.

Sentence starter: I found … [unit]. This value means … in this situation.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **0.1 kg** metal at **100°C** is placed in **0.2 kg** water at **20°C**. Final temperature is **25°C**; use water c=**4200** and neglect container/losses. **Find metal c**.

Sentence starter: I found … [unit]. This value means … in this situation.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3